

Yun-Tsz Tsai (Clara)

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SUMMARY

Data engineer focused on building reliable, governed data foundations: warehouse and pipeline design using medallion architecture (bronze/silver/gold), dbt-based transformation and testing on Google BigQuery, SQL-based data modeling, workflow orchestration (Apache Airflow), and REST APIs (Python/FastAPI). Experienced in identifying data quality issues, duplicated logic and unclear ownership, and turning them into documented, trustworthy, automatically-tested datasets — with daily use of Git, Docker, dbt, and CI/CD in Linux environments. Brings a multicultural, cross-functional communication background and a collaborative, detail-oriented approach to data ownership.

TECHNICAL SKILLS

Data Modeling & Warehousing: SQL, dbt, medallion architecture (bronze/silver/gold layering), star-schema style modeling, data warehousing (MySQL, Google BigQuery), ETL/ELT design

Data Quality & Governance: dbt tests (unique, not_null, relationships, custom singular tests), data quality & consistency validation, canonical entity / taxonomy design, documentation and naming conventions, data governance concepts (lineage, quality frameworks)

Orchestration & Engineering: Apache Airflow, REST API development (FastAPI), distributed ingestion architectures, CI/CD (GitHub Actions), Infrastructure as Code (Terraform), RabbitMQ, Docker, Git, Linux

Languages & Frameworks: Python (FastAPI, ETL/ELT, statistical analysis, Hugging Face transformers), SQL, Node.js

Cloud & BI: Google Cloud Platform (BigQuery, Google Cloud Engine), Redash, Power BI

PROJECTS

SQL Data Warehouse — Medallion Architecture (Bronze / Silver / Gold)

May 2026 – Aug 2026

- Designed and built a multi-layer data warehouse using medallion architecture, separating raw, cleansed, and business-ready data into distinct, clearly owned layers to keep transformation logic transparent and auditable
- Migrated the warehouse from MySQL to Google BigQuery, rebuilding transformation logic as dbt models and replacing manual validation with automated dbt tests (uniqueness, not-null, and referential integrity checks) across 9 models
- Implemented a CI/CD pipeline with GitHub Actions to automatically build and test the warehouse on every push, and generated auto-documented data lineage via dbt docs
- Brought existing GCP infrastructure (BigQuery datasets, service account, IAM roles) under Terraform management, using terraform import to reconcile manually-created resources with code and verifying zero configuration drift via terraform plan

Job Hunter — Distributed Data Ingestion Pipeline with REST API & Web Interface

Mar 2026 – Ongoing

- Designed and built a distributed ingestion pipeline collecting listings from two external platforms simultaneously, using a 3-worker distributed architecture to process 3,000+ records per run, orchestrated with Apache Airflow
- Structured and stored ingested data in a MySQL database; developed a REST API (Python/FastAPI) and a Node.js front end so end users can search and browse the dataset
- Developed and tested the full pipeline in a Linux/Docker environment; used Git for version control throughout

WordPress Auto-Categorization Pipeline

Jun 2026

- Built an automated ETL pipeline that ingests blog content, classifies it using NLP techniques, and syncs results back via the WordPress API — removing a manual categorization step and increasing categorization efficiency by 63%

Master's Thesis — Large-Scale Text Classification Pipeline

Feb 2026 – May 2026

- Built a scalable text-classification workflow using pre-trained language models to ingest and classify 19M+ online comments at reasonable runtime, applied to a research question on differential response patterns to men and women in online comments

EXPERIENCE

Data & Systems Intern — Beredskapslyftet, Remote

Dec 2025 – Apr 2026

- Implemented data quality and consistency checks across an HR data platform, identifying and correcting inaccurate or stale records to keep the system trustworthy for downstream reporting

Student Ambassador — Linköping University, Sweden

Oct 2025 – Jun 2026

- Conducted data analysis on enrollment trends, including year-over-year comparisons, to evaluate marketing effectiveness
- Produced marketing materials and supported ad-hoc, data-driven decision making

Research Assistant — Taiwan Institute of Economic Research (TIER), Taipei, TW

Sep 2022 – Dec 2022

- Analyzed market data and built visualizations in Excel to support investment research, including India market entry and renewable energy sectors

EDUCATION

M.Sc., Computational Social Science — Linköping University, Linköping, SE Aug 2024 – Jun 2026

Exchange Program — Linköping University, Faculty of Arts and Science, SE Jan 2023 – Jun 2023

Exchange Program — University of Tsukuba, School of Social and International Studies, JP Sep 2023 – Feb 2024

BA, Japanese Language and Literature — National Taiwan University (NTU), Taipei, TW — GPA 4.07 / 4.3 Sep 2019 – Jun 2024

CERTIFICATE

Google Cloud Data Analytics Issued Aug 2025

Hands-on experience with GCP tools including BigQuery and Looker Studio; covered ETL/ELT and data governance concepts.

LANGUAGES

Mandarin: Native | English: Fluent (IELTS 7.5) | Japanese: Fluent (JLPT N1)